

# Principles of Communication Systems

## Module 1.

### Random variables and processes.

- Introduction.
- Probability.
- Conditional Probability.
- Random variables.
- Statistical averages.
- Function of a random variable.
- Moments.
- Random process
- Mean.
- Correlation and covariance function.
- Properties of autocorrelation function.
- Properties of cross-correlation functions.
- Gaussian process.
- Gaussian Distribution Function.



# PRINCIPLES OF COMMUNICATION SYSTEMS -

## Module 1 - RANDOM VARIABLES & PROCESSES.

### Introduction:

- \* Random signals are present in every practical communication system.
- \* A signal is random, if it is not possible to predict its precise level in advance.
- \* Every speech signal is a burst of energy in random times.

Major source of receiver noise is thermal noise which is due to random motion of electrons.

- \* For a random signal detection of precise value is difficult. So we describe a statistical properties like Average power in random signal or average spectral distribution of power. This is done with a probability theory.
- \* Processing of random signal is a random process and it consists of sample function.



## PRINCIPLES OF COMMUNICATION SYSTEMS - (UNIT 1)

### Probability theory:-

\* It gives the chance of getting the event.  
An experiment is repeated and the outcome may differ due to random phenomenon or chance mechanism.

Example: Tossing a coin for 'n' trial,  
if we get head for 'm' times,  
then  $\frac{m}{n}$  is probability of Head.

Let us consider an experiment where

- 1) The set of all possible outcomes of experiment is the sample space 'S'
- 2) An event may be a single sample point or set of sample points.
- 3) A single sample point is an elementary event.
- 4) The entire sample space 'S' is sure event and null set  $\phi$  is impossible event.

Sample space 'S' may be discrete or continuous.

Example of Discrete events, Tossing coin or Dice.  
Continuous events are measurement of voltage or noise.



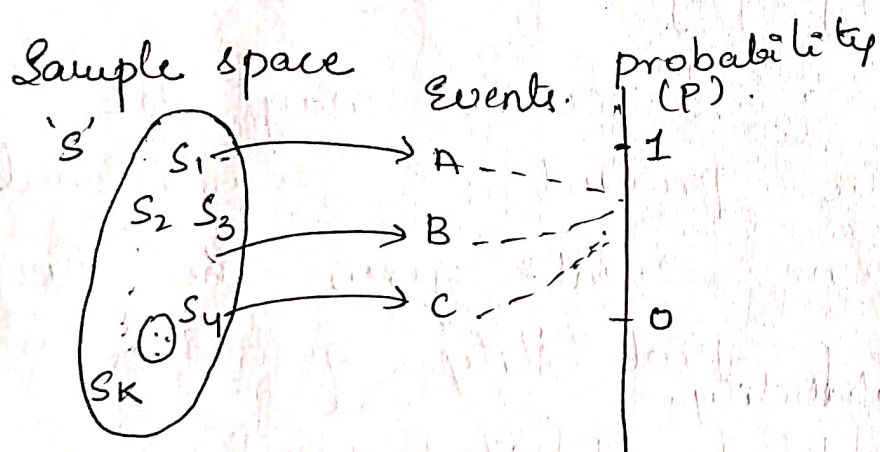
## PRINCIPLES OF COMMUNICATION SYSTEMS

### Mutually exclusive events:

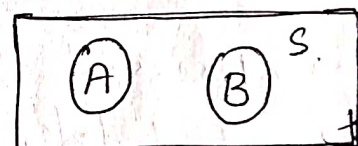
Two events are mutually exclusive, if occurrence of one event preclude (stops) the occurrence of other event.

### probability axioms (properties)

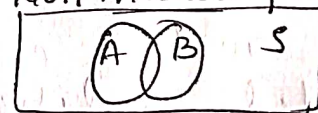
- 1) probability of any event will be b/w 0 & 1  
i.e.  $0 \leq P(A) \leq 1$ .
- 2)  $P[S] = 1$ ; i.e. Total probability of sample space is 1.
- 3) If 'A' & 'B' are 2 mutually exclusive events then  $P[A \cup B] = P[A] + P[B]$



Venn diagram of mutually exclusive events



Non mutually exclusive



If P is area of space 'S' with  $S \geq 1$ .

then 1.  $P[\bar{A}] = 1 - P[A]$ .

2.  $P[A \cup B] = P[A] + P[B] - P[A \cap B]$

↳ joint event

3)  $P[A_1] + P[A_2] + P[A_3] + \dots + P[A_m] = 1$ .



## Conditional probability.

Suppose in an experiment a pair of outputs 'A' & 'B' are events.

Let  $P[B/A]$  is probability of event 'B'

given that 'A' has occurred, then  $P[B/A]$  is a conditional probability.

$$\text{i.e. } P[B/A] = \frac{P[A \cap B]}{P[A]} ; \text{ where } P[A \cap B] \text{ is joint probability}$$

$$\therefore P[A \cap B] = P[B/A] P[A]$$

$$\text{or } P[A \cap B] = P[A/B] P[B]$$

Joint probability of 2 events may be expressed as the product of conditional probability of one event given the other and elementary probability of other. is called conditional probability.

$$\text{Baye's rule} = P[B/A] = \frac{P[A/B] \cdot P[B]}{P[A]}$$

if  $P[A] \neq 0$ .



(Random variable is a rule, which assigns a real number to each possible outcomes)

## PRINCIPLES OF COMMUNICATION SYSTEMS -

### Random variable:

A function whose Domain is a sample space and range is a set of real numbers is called as Random variable of the experiment.

Example: In Tossing a coin, Head corresponds to '1' and Tail to '0'.

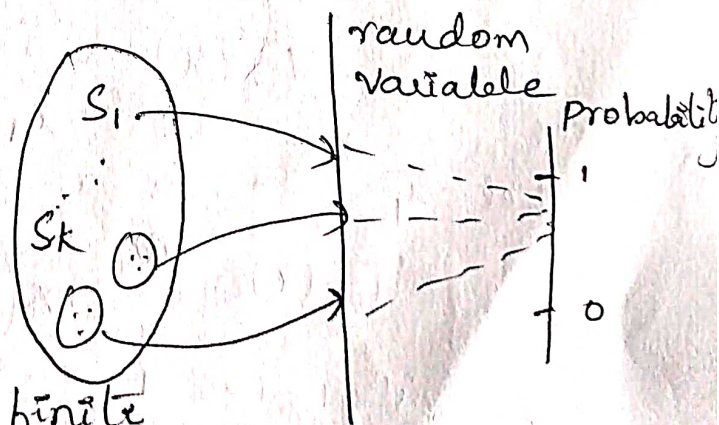
These numbers are assigned to outcomes of experiments.

∴ If the outcome of experiment is 's' then random variable is  $X(s)$  or  $x$ .

The particular outcome of random experiment is  $X(s_k) = x$ .

Note: There may be more than one random variable associated with the same random experiment

Random variable can take Discrete values or Continuous values.



If Random variable takes only finite number of discrete values, then it's called discrete random variable.

Rashmi M Hullamani | Assistant Professor, Department of Telecommunication, JNNCE Shivamogga

If a Random variable takes continuous values, then it is called continuous random variable. (3)



## PRINCIPLES OF COMMUNICATION SYSTEMS -

Let 'x' be a random variable.

probability of event ' $x \leq \alpha$ '

$$\text{ie } P[X \leq \alpha]$$

The function  $F_X(x) = P[X \leq \alpha]$  ~~is a~~

is a cumulative distribution function [CDF]

where 1)  $F_X(x)$  is bounded b/w zero and one.

2)  $F_X(x)$  is a monotone - non decreasing function;

$$\text{ie } F_X(x_1) \leq F_X(x_2) \quad \text{if } x_1 < x_2.$$

The derivative of CDF,  $F_X(x)$  is probability density function [PDF]

$$\text{ie PDF, } f_X(x) = \frac{dF_X(x)}{dx}$$

$$P[x_1 \leq X \leq x_2] = P[X \leq x_2] - P[X \leq x_1]$$

$$= F_X(x_2) - F_X(x_1)$$

$$F_X(x) = \int_{x_1}^{x_2} f_X(x) dx$$

$$\text{let } x_1 = -\infty \quad \& \quad x_2 = \infty$$

$$F_X(x) = \int_{-\infty}^{\infty} f_X(\xi) d\xi$$

$$\text{But } F_X(\infty) = 1 \quad \text{and} \quad F_X(-\infty) = 0.$$

$$\therefore \int_{-\infty}^{\infty} f_X(x) dx = [F_X(x)]_{-\infty}^{\infty}$$

$$= F[\infty] - F[-\infty]$$

$$= 1 - 0 = \underline{\underline{1}}$$



## PRINCIPLES OF COMMUNICATION SYSTEMS -

### Several Random Variables:

Few experiments requires several random variables for description.

So if 'X' & 'Y' are two random variable. Joint probability distribution is  $F_{X,Y}(x,y)$ .

where  $x, y$  are ~~sup~~ specified values.

Then o/p of experiment result is sample point lying inside

$(-\infty < x \leq x', -\infty < y \leq y')$  of joint space.

$$\text{i.e. } F_{X,Y}(x,y) = P[X \leq x, Y \leq y]$$

Suppose joint PDF is continuous then partial derivation

$$\text{pdf } f_{X,Y}(x,y) = \frac{\partial^2 F_{X,Y}(x,y)}{\partial x \partial y} \text{ is a continuous.}$$

Joint PDF is monotonic, Non decreasing function of both  $x$  and  $y$ .

The total values under graph of the joint probability density function must be unity as shown by

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(\xi, \eta) d\xi d\eta = 1.$$

## PRINCIPLES OF COMMUNICATION SYSTEMS -

The conditional probability density function of 'y' given that  $X=x$ , is defined by

$$f_y(y/x) = \frac{f_{x,y}(x,y)}{f_x(x)}$$

Statistical Average:

It is the expected value or mean of random variable 'x'

$$\text{i.e. } \mu_x = E[X] = \int_{-\infty}^{\infty} x f_x(x) dx.$$

Mean gives center of gravity of an area under probability density curve.

Joint probability density function:

- It is the PDF of two (or) more random variables.

- Joint PDF of two random variables x & y is given by the partial derivative of the joint distribution function.

$$f_{xy}(x,y) = \frac{\partial^2 F_{xy}(x,y)}{\partial x \partial y}.$$



Function of a Random variable:

Let 'x' is a random variable with real-valued function  $g(x)$ , then expected value of  $y$  with PDF  $f_Y(y)$

$$y = g(x).$$

$$E[Y] = \int_{-\infty}^{\infty} y \cdot f_Y(y) dy.$$

$$\underline{\underline{E[g(x)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx.}}$$

Q) The random variable  $y$  is the function of another random variable 'x' such that  $y = \cos(x)$ , where  $x$  is a random variable uniformly distributed in interval  $(-\pi, \pi)$ .

$$\text{ie } f_X(x) = \begin{cases} \frac{1}{2\pi} & -\pi < x < \pi \\ 0 & \text{otherwise} \end{cases}$$

Find out the mean of  $y$

Solution:

$$E[Y] = \int_{-\pi}^{\pi} \cos x \cdot \frac{1}{2\pi} dx.$$

$$= \frac{1}{2} \left[ \sin x \right]_{-\pi}^{\pi}$$

$$\underline{\underline{E[Y] = 0.}}$$

### Moments:

- Let 'X' be a random variable with a real valued function  $g(x) = x^n$  then  $n^{\text{th}}$  moment of probability distribution of X is

$$E[X^n] = \int_{-\infty}^{\infty} x^n f_X(x) dx.$$

First two moments are important.

for  $n=1$ ,  $E[X] = \int_{-\infty}^{\infty} x f_X(x) dx$  is a mean of random variable

for  $n=2$ ,  $E[X^2] = \int_{-\infty}^{\infty} x^2 f_X(x) dx$  is a mean square value of 'X'

### Central Moments:

It is moments of the difference b/w a random variable (X) and its mean  $\{\mu_X\}$

$\therefore n^{\text{th}}$  central moment is

$$E[(X - \mu_X)^n] = \int_{-\infty}^{\infty} (x - \mu_X)^n f_X(x) dx.$$

for  $n=1$ , central moment is zero.

for  $n=2$ , 2<sup>nd</sup> central moment is variance of random variable 'X'

$$\text{i.e. } \text{Var}[X] = E[(X - \mu_X)^2] = \int_{-\infty}^{\infty} (x - \mu_X)^2 f_X(x) dx.$$



# Principles of Communication Systems

- Variance is denoted as  $\sigma_x^2$   
Square root of the variance is standard deviation of random variable  $x$ .

$$SD = \sqrt{\text{Variance}}$$

$$= \sqrt{\sigma_x^2}$$

$$= \sigma_x$$

$$SD = \sqrt{E[x^2] - \mu_x^2}$$

## Random process:-

- \* when the random variable is a function of one independent variable (time) it is called as random process.
- \* It is denoted by  $x(t)$
- \* Random process or stochastic process is a sample space & ensemble

# Principles of Communication Systems

Difference b/w Random variable and Random process.

| Random variable                                | Random process                                           |
|------------------------------------------------|----------------------------------------------------------|
| 1) It is a set of numbers.                     | 1) It is a waveform.                                     |
| 2) It need not be function of time             | 2) It is the function of time.                           |
| 3) Random variable are not further classified. | 3) Random processes can be<br>- stationary<br>- Ergodic. |
| 4) only ensemble average can be calculated     | 4) Ensemble as well as time average can be calculated    |



## PRINCIPLES OF COMMUNICATION SYSTEMS -

\* Variance is denoted as  $\sigma_x^2$ .

\* Square root of the variance.

$\sqrt{\sigma_x^2} = \sigma_x$  is standard deviation of random variable 'X'.

$$SD = \sqrt{\text{Variance}}$$

$$= \sqrt{\sigma_x^2}$$

$$\equiv$$

$$= \sigma_x.$$

$$SD = \sqrt{E[x^2] - \mu_x^2}$$

Mean:-

For a Random process  $X(t)$ , the mean of the process  $X(t)$  as the expectation of the random variable obtained by observing the process at some time 't' is given by

$$\mu_x(t) = E[X(t)].$$

$$= \int_{-\infty}^{\infty} x f_{X(t)}(x) dx.$$

where  $f_{X(t)}(x)$  is PDF of process at time 't'.

Note: The mean of stationary process is constant i.e.  $\mu_x(t) = \mu_x$  for all t.

### Auto correlation:

The auto correlation function of the process  $X(t)$  is the expectation of the product of two random variables  $X(t_1)$  and  $X(t_2)$ , obtained by observing  $X(t)$  at times  $t_1$  &  $t_2$  respectively.

$$\begin{aligned} \text{i.e. } R_X(t_1, t_2) &= E[X(t_1)X(t_2)] \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_1 x_2 f_{X(t_1), X(t_2)}(x_1, x_2) dx_1 dx_2 \end{aligned}$$

where,  $f_{X(t_1), X(t_2)}(x_1, x_2)$  is a joint probability density of the random variable  $X(t_1)$  &  $X(t_2)$ .

The auto correlation function of a stationary random process depends only on the time difference  $t_2 - t_1$  as  $R_X(t_1, t_2) = R_X(t_2, t_1)$  for all  $t_1, t_2$ .

### Auto-covariance:

Auto-covariance function of a stationary random process  $X(t)$  is

$$\begin{aligned} C_X(t_1, t_2) &= E[(X(t_1) - \mu_X)(X(t_2) - \mu_X)] \\ &= R_X(t_2 - t_1) - \mu_X^2 \end{aligned}$$



Properties of Auto-Correlation function:

Defining the auto-correlation function of stationary process  $x(t)$  as

$$R_x(\tau) = E[x(t+\tau)x(t)] \text{ for all } t \text{ ——— (1)}$$

① The mean square value of the process may be obtained from  $R_x(\tau)$  by putting  $\tau=0$ .

$$\begin{aligned} \therefore R_x(0) &= E[x(t+0)x(t)] \\ &= E[x^2(t)] \end{aligned}$$

② The auto-correlation function  $R_x(\tau)$  is an even function of  $\tau$ .

$$\text{i.e. } R_x(\tau) = R_x(-\tau).$$

we can rewrite (1) as,

$$R_x(\tau) = E[x(t) \cdot x(t-\tau)]$$

③ The auto-correlation function  $R_x(\tau)$  has its maximum magnitude at  $\tau=0$ .

$$\text{i.e. } |R_x(\tau)| \leq R_x(0).$$

Proof: consider non negative quantity.

$$E[(x(t+\tau) \pm x(t))^2] \geq 0.$$

Expanding

$$E[x^2(t+\tau)] \pm 2 \cdot E[x(t+\tau)x(t)] + E[x^2(t)] \geq 0.$$

From property (1),

$$R_x(0) \pm 2 \cdot R_x(\tau) + R_x(0) \geq 0.$$

$$2R_x(0) \pm 2R_x(\tau) \geq 0.$$

$$-R_x(0) \leq R_x(\tau) \leq R_x(0)$$

Cross-correlation function:

Let us consider  $x(t)$  and  $y(t)$  are 2 random processes with auto-correlation function

$$R_x(t, u) \text{ and } R_y(t, u).$$

The cross correlation function of  $x(t)$  and  $y(t)$  is defined by.

$$R_{xy}(t, u) = E[x(t) \cdot y(u)]$$

and

$$R_{yx}(t, u) = E[y(t) \cdot x(u)]$$

If  $x(t)$  &  $y(t)$  are stationary then

$$R_{xy}(t, u) = R_{xy}(\tau)$$

$$\text{where } \tau = t - u.$$



## PRINCIPLES OF COMMUNICATION SYSTEMS -

### Properties of Cross-correlation:

- 1) Cross correlation function is not generally an even function of  $\tau$ .
- 2) It does not have maximum at the origin.
- 3) It obeys certain Symmetry  $R_{xy}(\tau) = R_{yx}(-\tau)$ .

Q) A random variable has PDF given by  $f_X(x) = 2e^{-2x}$  for  $x > 0$ . Find the probability that it will take a value b/w 1 & 3.

Solution:  $P(x_1 \leq x \leq x_2) = \int_{x_1}^{x_2} f_X(x) dx$

Here  $x_1 = 1$  &  $x_2 = 3$ .

$$P(1 \leq x \leq 3) = \int_1^3 2e^{-2x} dx.$$

$$= 2 \cdot \frac{e^{-2x}}{-2} \Big|_1^3$$

$$= -1 (e^{-6} - e^{-2})$$

$$= e^{-2} - e^{-6}$$

$$\boxed{P(1 \leq x \leq 3) = 0.1328}$$

## PRINCIPLES OF COMMUNICATION SYSTEMS -

Q) The PDF of a random variable is given as  $f_x(x) = \begin{cases} K & \text{for } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$

where  $K$  is constant.

i) Sketch the PDF & determine value of  $K$ .

ii) If  $a = -1$  &  $b = 2$ , calculate  $P(|x| \leq c)$  for  $c = \frac{1}{2}$

Solution:-

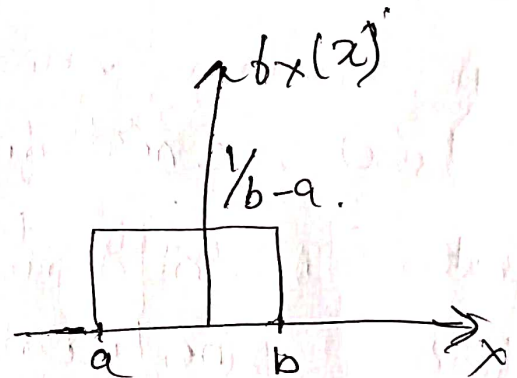
① w.k.t  $\int_{-\infty}^{\infty} f_x(x) dx = 1$

$$= \int_a^b K dx = 1$$

$$= K x \Big|_a^b = 1$$

$$K(b-a) = 1$$

$$\boxed{K = \frac{1}{b-a}}$$



∴ PDF can be expressed as

$$f_x(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$



# Principles of Communication Systems

ii) To determine  $P(|x| \leq c)$  for  $c = 1/2$ .

i.e.  $|x| \leq 1/2$ ,  $P(-1/2 \leq x \leq 1/2)$ .

$$P(x_1 < x < x_2) = \int_{x_1}^{x_2} f_x(x) dx.$$

$$P(-1/2 \leq x \leq 1/2) = \int_{-1/2}^{1/2} \frac{1}{b-a} dx$$

$$= \frac{1}{b-a} \left[ x \right]_{-1/2}^{1/2}$$

$$= \frac{1}{b-a} \left[ \frac{1}{2} - (-1/2) \right]$$

$$= \frac{1}{b-a} \left[ \frac{1}{2} + \frac{1}{2} \right]$$

$$P(-1/2 \leq x \leq 1/2) = \frac{1}{b-a}$$

Putting  $a = -1$ ,  $b = 2$ .

$$P(-1/2 \leq x \leq 1/2) = \frac{1}{2 - (-1)}$$

$$= \frac{1}{3}$$

# Principles of Communication Systems

## Gaussian process:

\* In Probability theory and statistics, a Gaussian process is a stochastic process (a collection of random variables indexed by time or space), such that every finite collection of those random variables has a multivariate normal distribution.

## Gaussian distribution function:

\* Gaussian distribution function is also known as normal distribution, is a probability distribution that is symmetric about the mean, showing that data near the mean are more frequent in occurrence than data far from the mean.

\* The graph of a Gaussian is a characteristic symmetric "bell curve" shape.



Total area = 1.



# Principles of Communication Systems

Properties of Gaussian PDF: (Probability distribution function)

Property 1:

\* The Gaussian PDF plot is symmetrical (even symmetry) about mean value ( $x=m$ ).  $x=m$  is also the 'media' and 'mode' of the distribution.

$$\Rightarrow f_x(m-\sigma) = f_x(m+\sigma).$$

Property 2:

\* The Peak of the Gaussian PDF is at its mean value i.e. at  $x=m$ .  
So we can calculate the peak value  $x=m$ .

$$\Rightarrow f_x(x) = \frac{1}{\sigma\sqrt{2\pi}} \quad \text{at } x=m.$$

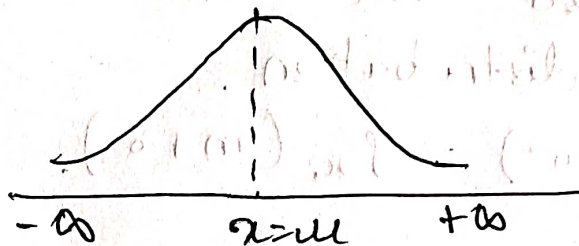
# Principles of Communication Systems

## Property 3:

\* The area under the Gaussian PDF curve below and above the mean value is  $\frac{1}{2}$

$$P(X \leq m) = P(X \geq m) = \frac{1}{2}$$

The area under the curve is 1.



Total area = 1  
 $-\infty$  to  $x \geq 0.5$   
 $x$  to  $\infty = 0.5$

## Property 4:

\* For Gaussian distribution plot, its first derivative is positive for  $x < m$ , equal to 0 at  $x = m$  & negative for  $x > m$ .